

Uniqueness of the Rank-5 Projection in the Quantum Gauge Theory of Time

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Abstract

We prove that $r = 5$ is the unique rank compatible with the foundational requirements of the QGT framework: Noetherian conservation, autosimilar closure, and operational metric stability. The proof is a two-sided exclusion with no free parameters.

Keywords. Non-invertible projection; rank uniqueness; autosimilar closure; boundary metric; Majorana spinor; finite-rank operator.

The QGT programme derives physical structure from the conservation law $\partial_I J^I = 0$ under restricted accessibility [1]. This produces a non-invertible projection $\Pi : \mathcal{M}^5 \rightarrow \mathbb{R}^4$. The admissible finite rank is constrained by three independent arguments across the corpus; this note proves that the only rank satisfying all of them is $r = 5$: spectral curvature gives $r \geq W = 5$ where $W = |\lambda_+ - \lambda_-|^2 = 5$ is the squared separation of the Fibonacci roots; holonomy stability gives $n = 5$ as the unique sector without collapse or redundancy [1]; metric stability gives $w_4 > \alpha > w_5$ where α is derived independently from the Pincherle operator and Fibonacci curvature, not from the rank [2]. Each of these establishes that $r = 5$ works. None proves, in isolation, that no other rank could also work. This note closes that gap.

Theorem 1 (Uniqueness of rank-5 projection). *Let $\Pi_r : \mathcal{M}^5 \rightarrow \mathbb{R}^4$ be a finite-rank non-invertible projection from a Majorana bulk with autosimilar structure φ , satisfying simultaneously:*

- (I) *Noetherian conservation $\partial_I J^I = 0$ with balance across accessible and coherent sectors;*
- (II) *second-order autosimilar closure $v_{k+2} = v_{k+1} + v_k$, forced by non-Abelian residual transport under restricted accessibility;*
- (III) *the induced boundary metric d_∂ is non-degenerate and operationally stable.*

Then $r = 5$ is the unique admissible rank.

Proof. $r \geq 5$ from (I) and (II). The autosimilar closure requires one resolvable singular direction per closure direction of the recursion; with $W = 5$ closure directions, $r \geq 5$ [2]. For $r \leq 3$, the minimal non-Abelian structure $U(1) \times SU(2)$ cannot be realised and (II) fails. For $r = 4$, the fifth closure step ($k = 4$) has no representative in the retained subspace; the Noetherian balance (I) cannot be maintained. Hence $r \geq 5$.

$r \leq 5$ from (III). The threshold α is derived in [2] from the Pincherle scale increment $2\varphi^2$ and Fibonacci curvature $\sqrt{5}$, independently of any rank value. The transition weights $w_k = \sigma_k \sigma_{k+1} = \varphi^{-(2k+1)}$ satisfy $w_4 > \alpha > w_5$ as a verified consequence. For $r > 5$, mode $k = 5$ has $w_5 < \alpha$: including it assigns positive metric weight to an operationally invisible direction,

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violating (III). By the holonomy argument, modes $k \geq 5$ also generate no new independent bilinear invariants, introducing redundant metric directions. Hence $r \leq 5$.

Therefore $r = 5$, satisfying all three requirements. $\square$ $\square$

Corollary 2. *Under the hypotheses of Theorem 1, in the SVD representative of [2]: the retained singular values are $\sigma_k = \varphi^{-k}$, $k = 0, \dots, 4$; the boundary metric is $d_{\partial}^2 = \sum_{k=0}^4 \varphi^{-2k} |c_k|^2$; the threshold α satisfies $w_4 > \alpha > w_5$; in the QGT angular readout convention, $\mathcal{N}_{\text{adm}} = \{2, 3, 5, 7, 11\}$.*

Scope. The theorem takes as inputs the Majorana condition and the autosimilar structure φ ; it does not re-derive them. Every downstream result whose only open dependence was the uniqueness of $r = 5$ is closed, conditionally within the Majorana–Fibonacci projective class and the metric admissibility of [2]. The gauge group $G = U(1) \times SU(2) \times SU(3)$, the mixing angle $\sin^2 \theta_W = 3/13$, and the physical fine-structure constant α_{phys} were all derived under $r = 5$: the theorem shows this was not a free choice. Three open problems remain unaffected: the CMB transfer function $G_{\text{edge}}^{\eta\xi}$, the Yukawa map, and the $e\pi$ correction.

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References

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